ISEP INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

Escala

Fire Station Shift Management Platform

Sprint 1 – Idea Presentation and Progress Status Report

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1 Problem and Opportunity

1.1 Problem Context

The station's weekly schedule lives on paper. The supervisor defines the shifts and tracks, by hand, the minimum and maximum weekly hours of each member. The schedule is then edited in Excel and printed, and some members post a photo of it in the station's WhatsApp group.

Every change to the schedule means editing and reprinting it. This process causes three main problems:

- **Checking the schedule:** professional firefighters, volunteers and cadets have to go to the station to check their current schedule, or depend on other members posting updates in the WhatsApp group. Not every update is posted, so members often rely on an outdated schedule.
- **Keeping track of changes:** it is hard for the supervisor to follow every change, especially the shift extensions caused by emergencies and the swaps that members request. Some extensions and swaps are lost along the way, or never reach him at all.
- **Requesting swaps:** volunteers and cadets have to fit their shifts around their jobs, school and personal lives. To swap a shift, they must find another member to swap with and report the swap to the supervisor. Often only one of them reports it, and the other has no easy way of knowing whether the change was approved.

Together, these problems have led to a high turnover of volunteers and cadets.

1.2 Opportunity

There is an opportunity to implement **Escala**, a shift management app built for the station, whose need was raised by the cadets, volunteers and former members themselves. Escala addresses each of the problems above:

- **Checking the schedule:** a single, always up-to-date schedule, available on members' phones and on the station computers, replaces the printed sheet and the WhatsApp photos. Everyone is notified of the changes that affect them, so nobody has to go to the station or wait for an update that may never be posted.
- **Keeping track of changes:** every change is made and recorded in the app. The supervisor builds the weekly schedule with each member's availability, hours and limits in view, and publishes changes without reprinting. Professional firefighters register their extensions and call-outs themselves after the emergency, for the supervisor to review and approve, so no hours are lost.
- **Requesting swaps:** members mark their availability in advance and request swaps among themselves in the app. Both members see the result straight away, and the swap is logged for the supervisor.

Escala also gives dispatchers an on-shift board showing who is on shift at any moment, alongside the station's existing dispatch app. Since senior staff have little experience with technology, it is designed to be used from day one, without training. By making the schedule easier to follow and to fit around members' lives, Escala aims to reduce the turnover of volunteers and cadets.

1.3 Target Community

The target community is a Portuguese fire brigade association (*Associação Humanitária de Bombeiros*). One of its stations will be the pilot, with development funded by the municipality. Later, Escala can reach other fire brigades and humanitarian organisations that rely on volunteer shifts.

Within the station, the groups affected by the current schedule, and their problems, are listed in Table 1.

GROUP	PROBLEMS
Professional firefighters	• Emergencies extend shifts or cause night call-outs • Extensions can't be reported during the emergency, and the supervisor has often left by the time they return • Extra hours are miscounted or lost • The weekly day off is only known when the schedule is posted
Volunteers and cadets	• Limited, irregular availability • To know the final schedule they must go to the station or wait for a WhatsApp photo • Not every update is posted on WhatsApp • The schedule clashes with school, jobs and personal life
Supervisor	• Builds the schedule in Excel • Tracks each member's minimum hours and weekly limit by hand • Every change means editing and reprinting the schedule • Swaps and extensions made after hours, and the paper sheet for extensions, are hard to track and often go missing
Station chief	• Hard to know who is on shift at a given moment • Hard to find times when enough cadets and volunteers can attend training
Dispatchers	• The dispatch app shows who *has been dispatched*, not who is *on shift* • They have to rely on the Excel list to guess who is on shift

Table 1: Groups affected by the current schedule and their problems

Note: Senior staff, such as the supervisor and the station chief, have little familiarity with technology. Any tool must therefore be simple enough for them to adopt quickly and become part of the station's daily operations from day one, without training.

1.4 Product Vision Board

PRODUCT VISION BOARD

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VISION

What is the reason for creating the product? What positive change should it create?

Volunteers can organise their service around their lives and keep serving, so the corps has enough people when an emergency needs them.

TARGET GROUP

Which market or market segment does the product address? Who are the target customers and users?

Market: Portuguese fire brigade associations, starting with a pilot station funded by the municipality. Later, other humanitarian organizations that rely on volunteers shifts.

Customer: Fire brigade association.

Users:

- **Volunteers and cadets:** support the crews with limited and irregular availability.
- **Supervisors**: build and publish the weekly schedule around members' unavailability, and are informed of every swap and cover;
- **Shift Operators**: dispatch crews and need to know who is on shift;
- **Station Chief:** needs an overview of who is available to find cover, organise training and respond to emergencies;
- **Professional firefighters:** have fixed shifts that are often extended due to emergencies;

NEEDS

What problem does the product solve or which benefit does it offer? If you identify several needs, prioritise them and move the most important one to the top.

1) **Report availability and schedule shifts around it.** Volunteers and cadets report when they can't come, so they stop getting shifts that clash with work, studies or other commitments.
2) **Check and manage one's own schedule** without going to the station. Everyone can see the up-to-date schedule, and volunteers and cadets can negotiate swaps among themselves.
3) **Have enough people in emergencies and high fire risk.** When someone can't come, only free and qualified colleagues are asked to cover.
4) **Build the weekly schedule quickly and without conflicts**. Supervisors see members' unavailability and the required functions while they build it.
5) **An easy-to-use tool** that can be integrated into the daily operations without training.

PRODUCT

What product is it? What are its three to five stand-out features that set it apart from competing offering? Is it feasible to develop the product?

Type: A mobile web app (PWA) for volunteers, cadets and professional firefighters, also usable on the station computers.

Stand-out Features:

- **Unavailability in own words:** AI reads the dates and the member confirms them; a form is always available;
- **Self-service shift swaps** between members, with a function check;
- **Centralised up-to-date live schedule** with notifications;
- **Presence confirmation and cover requests**, sent only to free and qualified colleagues;

Feasibility: Standard web technology with no integration needed with the current dispatch app in use. Built incrementally in Sprints.

BUSINESS GOALS

How will the product benefit the company that develops and provides it? What are the desired business benefits? Prioritise them and move the most important one to the top.

1) **Reduce turnover among volunteers and cadets at the pilot stations,** so the corps can meet the minimum crew required in emergencies;
2) **Replace the printed Excel schedule and WhatsApp** with one live schedule
3) **Recurring revenue from selling the application to other fire brigades and humanitarian organizations** that rely on volunteers.
4) Additional revenue from **maintenance contracts** and **paid custom features**.

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Figure 1: Product Vision Board

1.5 Stakeholder Map

STAKEHOLDER	RELATION	INTEREST	INFLUENCE	ENGAGEMENT
Supervisor	Primary user: builds the schedule	High	High	Manage closely
Shift chief (*chefe de piquete*), from Firefighter 1st class up	Approves swaps, sees who is present before the shift and decides reinforcements	High	Medium	Manage closely
Station chief	Decision maker on adoption (decides whether the corps adopts the app)	Medium	High	Keep satisfied
Volunteers and cadets	Users	High	Medium	Involve in tests
Professional firefighters	Users	High	Medium	Involve in tests
Dispatchers	See who is on duty (read-only)	Medium	Low	Keep informed
Fire brigade association	Customer	Medium	High	Keep satisfied
Municipality	Possible funder (assumption)	Low	High	Monitor
Existing dispatch app	Neighbouring system, no integration	Low	Low	Monitor
IPMA	External data source (fire risk)	Low	Low	Monitor

Table 2: Stakeholder map

2 Personas

2.1 Target Customers and Users

Customer:

- Fire brigade associations (*Associações Humanitárias de Bombeiros*)
- Starting with the pilot station, whose development is funded by the municipality

Users:

- Supervisors
- Dispatchers
- The station chief
- Professional firefighters
- Volunteers and cadets

2.2 The Personas

Each persona stands for one of the target user groups. Table 3 gives a brief summary of the user personas: who they are, their main current problems and what they want solved. The full persona templates follow.

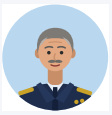
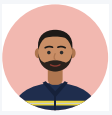
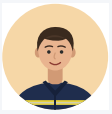
PERSONA	CURRENT PROBLEMS	WANTS SOLVED
 **Bernardino Santos** Supervisor, 59 *Supervisors*	• Every change means editing and reprinting the Excel schedule • Changes and extensions made after he leaves get lost • Tracks minimum hours and weekly limits by hand	• See availability, hours left and weekly limits • Update the schedule without reprinting it • Review and approve extensions registered in the app
 **Rui Carvalho** Station chief, 52 *Station command*	• Hard to find times when enough people can attend training • No quick overview of who is on shift • Couldn't learn Google Calendar or Notion	• Fewer volunteers and cadets quitting • See at a glance who is on shift and available • Plan training around known availability
 **Pedro Pacheco** Professional firefighter, 31 *Professional firefighters*	• Extra hours are hard to report and often go missing • Only learns his day off on Monday, at the station • Confirming a swap means going to the station or relying on WhatsApp	• Register every extra hour himself • Know his schedule and day off in advance • Check schedules on his phone when asked to swap
 **Diogo Ribeiro** Volunteer, 23 *Volunteers and cadets*	• Depends on WhatsApp to know about changes • Misses updates that are never posted • Station shifts clash with his Red Cross Sundays	• Know his schedule at any moment, on his phone • Set availability around his job and Red Cross shifts • Swap shifts and see the update straight away
 **Rita Machado** Former volunteer, 25 *Volunteers who quit*	• Irregular ICU shifts with frequent changes • Relied on colleagues to report her swaps • Was scheduled on top of her ICU shifts	• Mark when she isn't available, and update it easily • Request swaps from her phone and see them confirmed

Table 3: Brief summary of the user personas

2.3 Persona Templates

PICTURE & NAME	DETAILS	GOAL
What does the persona look like? How is it called? Choose a realistic and believable picture and name.  *Illustration created with AI* **Bernardino Santos**	What are the persona's relevant characteristics? For instance, demographics such as age, gender, occupation, and income; psychographics including lifestyle, social class, and personality; behavioural attributes like usage patterns, attitudes, and brand loyalty. Only state details that are relevant. **Demographics**: • Male, 59 years old, shift supervisor; • Former firefighter of 36 years. Stopped going out as a firefighter 4 years ago and now focuses on managing the station, which he already did before. **Psychographics**: • Manages the station: builds the weekly schedule, makes changes to it, and registers firefighters' extensions and night call-outs; • Was previously the station second-in-command and still uses his experience to help the station chief report to the municipality; • Helps answer calls and dispatch crews when he has time or when needed; • Has built the schedule in Excel for years with no difficulty, but struggles to learn new tools; • Methodical and responsible, and proud of keeping the station running. **Frustrations**: • Every change means updating the Excel sheet and the individual schedules, then printing them again; • Changes made after he leaves and extensions sometimes get lost or never reach him among all the work at the station; • There is a sheet to register extensions and call-outs, but it often goes missing; • Tracks the minimum hours and weekly limit of each member manually, and sometimes assigns more or fewer hours than expected, especially to new cadets and volunteers.	What problem does the persona want to solve or which benefit does the character want to gain? Why would the persona use or buy the product? If you identify more than one goal, prioritise them and state the main goal at the top. **Problems persona wants solved:** 1) See the availability of each member, the hours they still have to do, and their minimum and maximum weekly hours; 2) Build and update the weekly schedule quickly, without reprinting it every time something changes; 3) Let firefighters register their extensions and night call-outs in the app, so nothing gets lost, and review and approve them himself; 4) Know who is on shift at any moment when he helps with calls and dispatching crews. **Why would the persona buy the product?** • Bernardino has used Excel for years and doesn't want to learn a complex new tool, so the app must be simple enough to use from day one. He wants firefighters to register their own extensions and call-outs, but with him reviewing and approving each one, as letting anyone add hours without verification would make the hours count unreliable. With hours counted automatically and nothing lost on paper, he spends less time correcting the schedule.

Figure 2: Persona template – Bernardino Santos, supervisor

PICTURE & NAME	DETAILS	GOAL
What does the persona look like? How is it called? Choose a realistic and believable picture and name.  *Illustration created with AI* **Rui Carvalho**	What are the persona's relevant characteristics? For instance, demographics such as age, gender, occupation, and income; psychographics including lifestyle, social class, and personality; behavioural attributes like usage patterns, attitudes, and brand loyalty. Only state details that are relevant. **Demographics**: • Male, 52 years old, station chief, in the corps for 24 years. **Psychographics**: • Organises training for the whole station, but mainly trains the next generation of cadets and volunteers; • Only gets dispatched when no one else is available or for major emergencies; • Uses his phone for calls, messages and photos. Not in the volunteers' WhatsApp group and relies on the printed schedule; • Trains volunteers and cadets, so the responsibility of teaching them how to use the schedule app would also fall on him. Not used to complex apps and has little patience to learn them; • Experienced and respected. Focused on training volunteers and cadets, so he can one day leave the station to the current firefighters and the next generation with a clear mind. **Frustrations**: • Hard to find a time when enough cadets and volunteers are available for training; • No quick overview of who is on shift or available, so he checks the printed schedule or asks the supervisor; • Saddened to see cadets and volunteers who want to make a difference quit, because the way the schedule is handled doesn't let them organise themselves; • He and the supervisor tried Google Calendar and Notion. Most cadets and volunteers picked them up well, but the two couldn't learn them from day one, and can't risk failures in dispatch and personnel management.	What problem does the persona want to solve or which benefit does the character want to gain? Why would the persona use or buy the product? If you identify more than one goal, prioritise them and state the main goal at the top. **Problems persona wants solved:** 1) Reduce the number of cadets and volunteers who quit because of schedule conflicts, when they would rather stay and keep making a difference; 2) See at a glance who is on shift and who is available, without asking the supervisor or checking the printed schedule; 3) Know the availability of cadets and volunteers in advance, so he can plan training sessions when enough of them can attend. **Why would the persona buy the product?** • Rui wants to help the cadets and volunteers who leave only because they can't organise their schedules. A simple app that he can learn and introduce to them from day one, without risking failures in dispatch, lets them manage their own availability and shifts, and shows him who is on shift and who can attend training.

Figure 3: Persona template – Rui Carvalho, station chief

PICTURE & NAME	DETAILS	GOAL
What does the persona look like? How is it called? Choose a realistic and believable picture and name.  *Illustration created with AI* **Pedro Pacheco**	What are the persona's relevant characteristics? For instance, demographics such as age, gender, occupation, and income; psychographics including lifestyle, social class, and personality; behavioural attributes like usage patterns, attitudes, and brand loyalty. Only state details that are relevant. **Demographics**: • Male, 31 years old, professional firefighter for almost 8 years. **Psychographics**: • Works a fixed schedule from 8:00 to 17:00, including Saturdays; • His weekly day off changes from week to week; • Lives close to the station and made himself available to be called whenever needed. Often, when an emergency happens at night, he wakes up to the siren and goes to the station to help; • Comfortable with apps and uses his smartphone regularly; • Always ready to help when needed. **Frustrations**: • Extra hours, from shift extensions to night calls outside his shifts, are hard to report. By the time he gets back, the supervisor has often left and extra hours end up miscounted or go missing; • The schedule is only made at the end of each week, so he only finds out about his day off on Monday when he arrives at the station. Especially frustrating when his day off is that same Monday; • Whenever someone asks him to swap a shift or a day off, he has to go to the station to confirm the change or depend on someone posting the updated general schedule on WhatsApp.	What problem does the persona want to solve or which benefit does the character want to gain? Why would the persona use or buy the product? If you identify more than one goal, prioritise them and state the main goal at the top. **Problems persona wants solved:** 1) Have every hour he works outside his fixed schedule counted correctly, including shift extensions during emergencies or extra hours on night call-outs. He needs a way to register these hours himself, without depending on the supervisor or on colleagues' reports. 2) Know his schedule in advance, especially his day off, so he doesn't go to the station on a Monday that turns out to be his day off. 3) Be able to check his individual schedule and general schedule on his phone whenever someone asks him to swap a shift. **Why would the persona buy the product?** • Pedro already uses his smartphone for almost everything. Registering extra hours on his phone right after an emergency is far easier than reporting them to the supervisor. Additionally, he can also check his up-to-date schedule whenever someone asks him to swap a shift or day off, and no longer wastes a trip to the station when his day off turns out to be Monday.

Figure 4: Persona template – Pedro Pacheco, professional firefighter

PICTURE & NAME	DETAILS	GOAL
What does the persona look like? How is it called? Choose a realistic and believable picture and name.  *Illustration created with AI* **Diogo Ribeiro**	What are the persona's relevant characteristics? For instance, demographics such as age, gender, occupation, and income; psychographics including lifestyle, social class, and personality; behavioural attributes like usage patterns, attitudes, and brand loyalty. Only state details that are relevant. **Demographics**: • Male, 23 years old, volunteer firefighter for 3 years now. Joined as a cadet in middle school and now mostly drives the station's vehicles; • Works at an elderly care home, helping with entertainment and stimulating games and activities for residents. **Psychographics**: • Volunteers at the station after work on Thursdays and Fridays, and also on Saturdays and Sundays; • Additionally, he also volunteers at the Portuguese Red Cross, where his shifts rarely change; • Sometimes he needs to be at the care home at night for special occasions and celebrations; • Comfortable with apps and uses his smartphone regularly; • Caring and committed to helping others, but has very limited free time between his job and two volunteering commitments. **Frustrations**: • Doesn't have the time to drive to the station to confirm his schedule whenever changes occur, so he depends mostly on WhatsApp; • Sometimes misses updates on WhatsApp, or the update never gets posted, so he gets confused about his shifts; • The station sometimes schedules his shift to the same time as his Red Cross shifts on Sundays, creating conflicts; • Available to swap shifts with colleagues, but usually isn't the one requesting/reporting the swap, so he can't easily confirm the updated schedule.	What problem does the persona want to solve or which benefit does the character want to gain? Why would the persona use or buy the product? If you identify more than one goal, prioritise them and state the main goal at the top. **Problems persona wants solved:** 1) Know his schedule at any moment, on his phone, so he can plan his personal life and his Red Cross volunteering hours; 2) Be able to define his availability in advance, including the evenings he has special events at the care home and the shift hours he already has accepted at the Red Cross; 3) Swap shifts with others seamlessly and see the updated schedule right after the change. **Why would the persona buy the product?** • Diogo doesn't have time to drive to the station whenever a change affects his schedule. With an always up-to-date schedule on his phone, he can fit his volunteering at the station around his job and his Red Cross shifts, mark when he isn't available, and swap shifts with colleagues without having to drive to the station to check the updated schedule.

Figure 5: Persona template – Diogo Ribeiro, volunteer

PICTURE & NAME	DETAILS	GOAL
What does the persona look like? How is it called? Choose a realistic and believable picture and name.  *Illustration created with AI* **Rita Machado**	What are the persona's relevant characteristics? For instance, demographics such as age, gender, occupation, and income; psychographics including lifestyle, social class, and personality; behavioural attributes like usage patterns, attitudes, and brand loyalty. Only state details that are relevant. **Demographics**: • Female, 25 years old, ICU nurse; • Worked as a nurse at a health centre from 8:00 to 16:00 until July, then she moved to Intensive Care (ICU), where she now works in shifts. **Psychographics**: • Volunteered at the station for 2 years, 4 times a week, after work or on Saturdays, and sometimes on Sunday afternoons; • Quit after changing jobs, because of schedule issues; • Now volunteers at the Red Cross on Sundays, as the hours are consistent; • Comfortable with apps, uses her phone for almost everything; • Caring and committed to her community. Joined the Red Cross right after leaving, because she wants to make a difference; • Knows she can no longer volunteer 4 days a week plus Sundays, but wants a compromise to still help at the station, even if less than before. **Frustrations**: • No fixed ICU schedule, with frequent extensions and swaps; • Had no time to go to the station to report swaps or check her schedule. When colleagues accepted a swap, she had to rely on them to report it; • After she started at the ICU, the station often scheduled her shifts on top of her ICU shifts; • The ICU is far more exhausting. After two shifts on the same day (morning and afternoon), she may be too exhausted to volunteer, even if she is free.	What problem does the persona want to solve or which benefit does the character want to gain? Why would the persona use or buy the product? If you identify more than one goal, prioritise them and state the main goal at the top. **Problems persona wants solved:** 1) Define when she isn't available, especially on days with two consecutive ICU shifts, and update it whenever her ICU schedule changes, so the station doesn't schedule her when she can't go; 2) Request shift swaps efficiently from her phone and see them confirmed in the schedule, without going to the station or relying on colleagues to report them. **Why would the persona buy the product?** • Rita knows she can no longer volunteer as often as before, but she still wants to help. Marking herself unavailable on demanding ICU days and swapping shifts from her phone would let her find a compromise with the station: volunteering less than before, but without it clashing with her job, her rest or her Red Cross Sundays.

Figure 6: Persona template – Rita Machado, former volunteer

3 Alternatives

The station has already tried Google Calendar and Notion to manage the schedule, and dropped both because they took too long to learn.

ASPECT	GOOGLE CALENDAR AND NOTION
Learning curve	Took too long to learn and required some technical skill. Senior staff have little experience with technology, and there is no time for training.
Shift swaps	No shift-swap workflow: every swap would still have to be reported to the supervisor.
Who is on shift	No view of who is on shift at a given moment.
Hours	No way to track each member's minimum hours and weekly limit while building the schedule.

Table 4: Why the tools the station tried failed

These tools were not rejected for missing features alone: the station gave up on them because senior staff could not learn them quickly enough, and there is no time to train every member. Simplicity is therefore the deciding factor for adoption. Whatever tool the station uses must work from the first use, for every member, without training.

4 Value Proposition

4.1 What Escala Must Fix

For fire stations that mix professional firefighters, volunteers and cadets and run their schedule on a printed Excel sheet, Escala is a shared shift schedule that anyone can use from day one. It always shows who is on shift and lets members manage their availability, swaps and extensions themselves. Unlike generic scheduling apps or emergency-services tools designed for other markets, it needs no training and fits how Portuguese stations actually work.

For the station, the value lies in time and reliability. Table 5 shows what Escala changes for each group.

USER GROUP	VALUE OF ESCALA
Supervisor	• Stops rebuilding and reprinting the schedule • Sees each member's hours and limits without counting them by hand
Dispatchers	• Know at any moment who they can send
Professional firefighters	• No extra hours are lost after an emergency
Volunteers and cadets	• No more trips to the station to check the schedule • No more missed WhatsApp updates, which today lead some of them to give up volunteering

Table 5: Value of Escala for each user group

The pilot station uses Escala at no cost, as its development is funded by the municipality. Other institutions pay a subscription per active user, kept low enough for non-profit associations (indicatively €1–3 per user per month).

4.2 Main Features

USER GROUP	MAIN FEATURES
Professional firefighters	• Report shift extensions and off-hours shifts (call-outs) after the emergency, for the supervisor to approve
Professional firefighters, volunteers and cadets	• Define their unavailability in advance • Check their schedule on the phone and be notified of every change that affects them • Request and negotiate shift swaps among themselves
Supervisor	• Build the weekly schedule with each member's availability, minimum hours and weekly limit in view • Automatic checks for unavailability, unfilled roles, overlapping shifts and hours above the weekly limit • Publish the schedule, and any later change, without reprinting it • Review, approve or reject extensions and off-hours shifts • See every change, swap and extension in the change log
Station chief	• See who is on shift and who is available, to plan training sessions
Dispatchers	• On-shift board showing who is on shift at any moment, ready for dispatch • Complements the existing dispatch app, without integrating with it

Table 6: Main features of Escala for each user group

4.3 What Is Essential

Beyond its features, the following points are essential for Escala to be adopted by the station, as summarised in Table 7.

ESSENTIAL	DESCRIPTION
Usable from the first use	• Senior members of the station must be able to adapt quickly • Simple screens, large buttons, few steps, in Portuguese • No training needed
One central schedule	• Always up to date, with real-time updates • The people affected are notified of every change
Weekly hours tracking	• Minimum hours and weekly limit of each member, counted automatically

Table 7: What is essential for Escala to be adopted

5 User Journeys

Taking all of the above into account, five principal user journeys were defined, as summarised in Table 8.

JOURNEY	ACTORS	DESCRIPTION
UJ1 Plan the weekly schedule	• Professional firefighters, volunteers and cadets • Supervisor	• Members define their unavailability for the next week • The supervisor defines the shifts and the roles each one needs, and fills the schedule manually or automatically • Escala checks for conflicts before the schedule is approved and published • Every member is notified
UJ2 Register a shift extension or off-hours shift	• Professional firefighter • Supervisor	• After an emergency, the firefighter registers the extra hours • The supervisor approves them, or rejects them with a reason • The firefighter can ask for one more review • Approved hours update the schedule
UJ3 Ask for a shift swap	• Requesting member • Colleague • Supervisor (informed only)	• A member asks a specific colleague to swap, or publishes the request for anyone • Escala checks that each member can fill the other's role and stays within the weekly limit • Valid swaps update both schedules and appear in the supervisor's change log
UJ4 See who is on shift to dispatch	• Dispatcher	• The dispatcher opens the on-shift board • Sees who is on shift, with their role and shift end time • Dispatches the crew in the existing dispatch app
UJ5 Plan a training session	• Station chief • Selected members	• The station chief picks a shift and the members to train • Escala checks that key roles stay covered on shift • The selected members are notified

Table 8: Principal user journeys

5.1 User Journey Diagrams

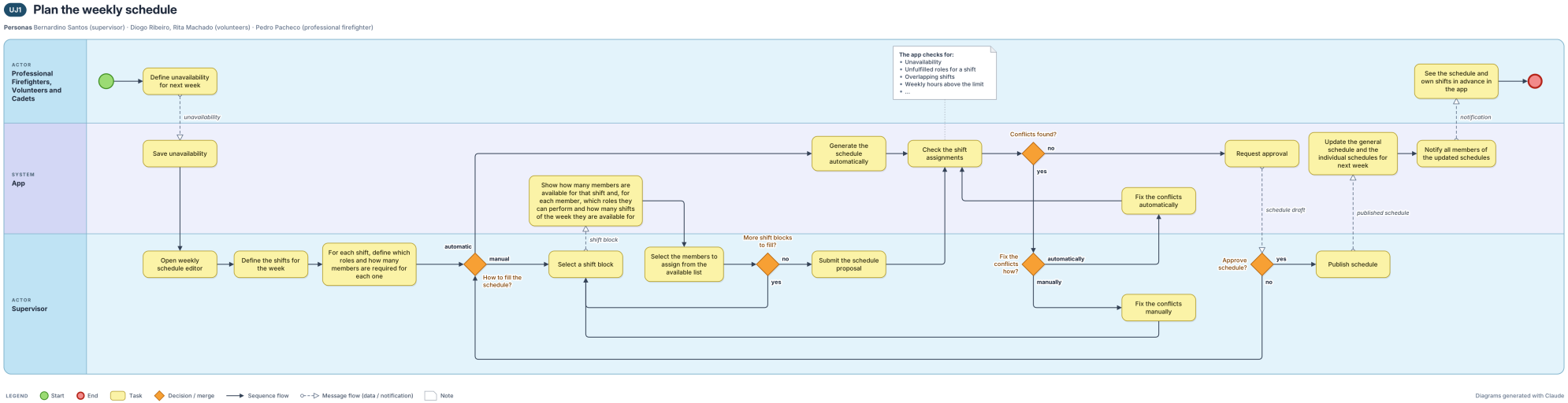


Figure 7: UJ1 – Plan the weekly schedule

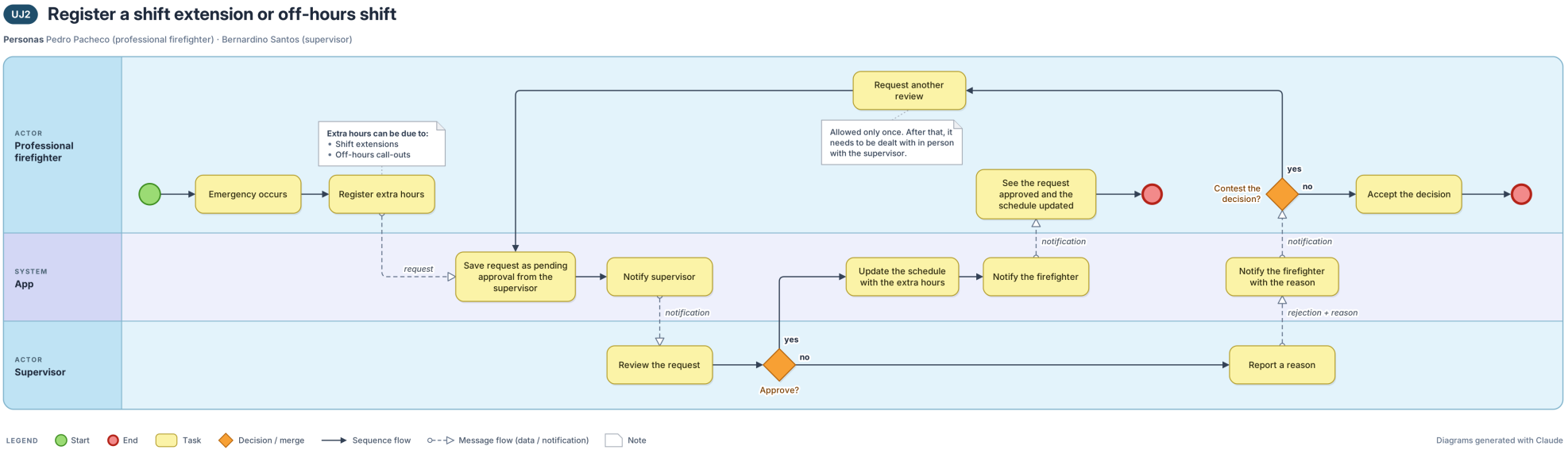


Figure 8: UJ2 – Register a shift extension or off-hours shift

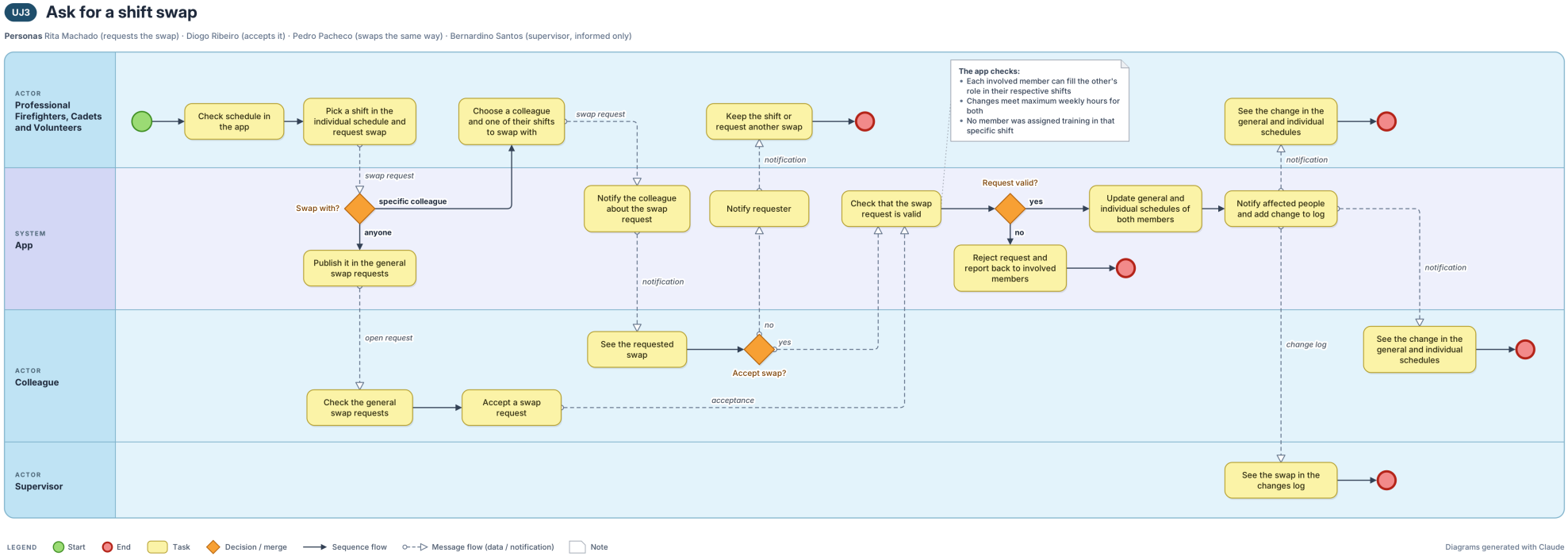


Figure 9: UJ3 – Ask for a shift swap

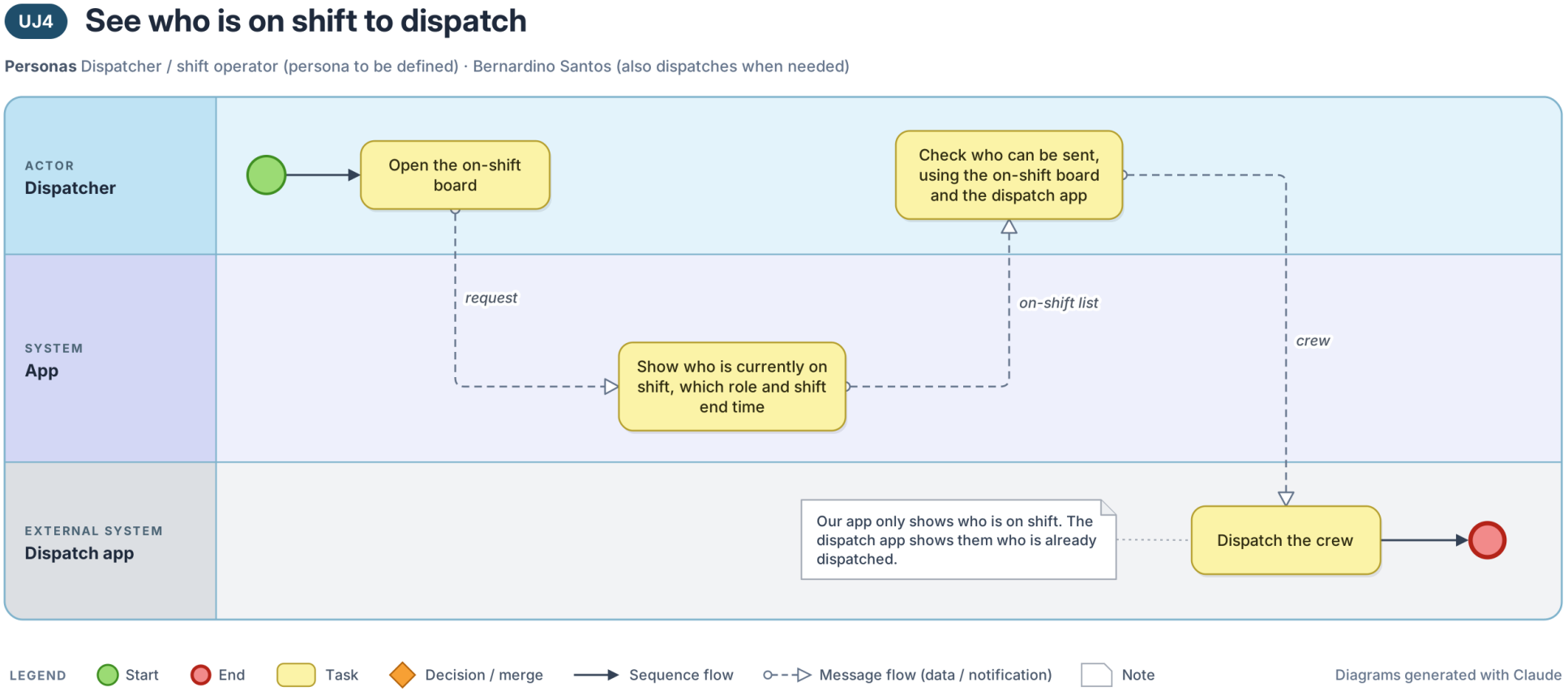


Figure 10: UJ4 – See who is on shift to dispatch

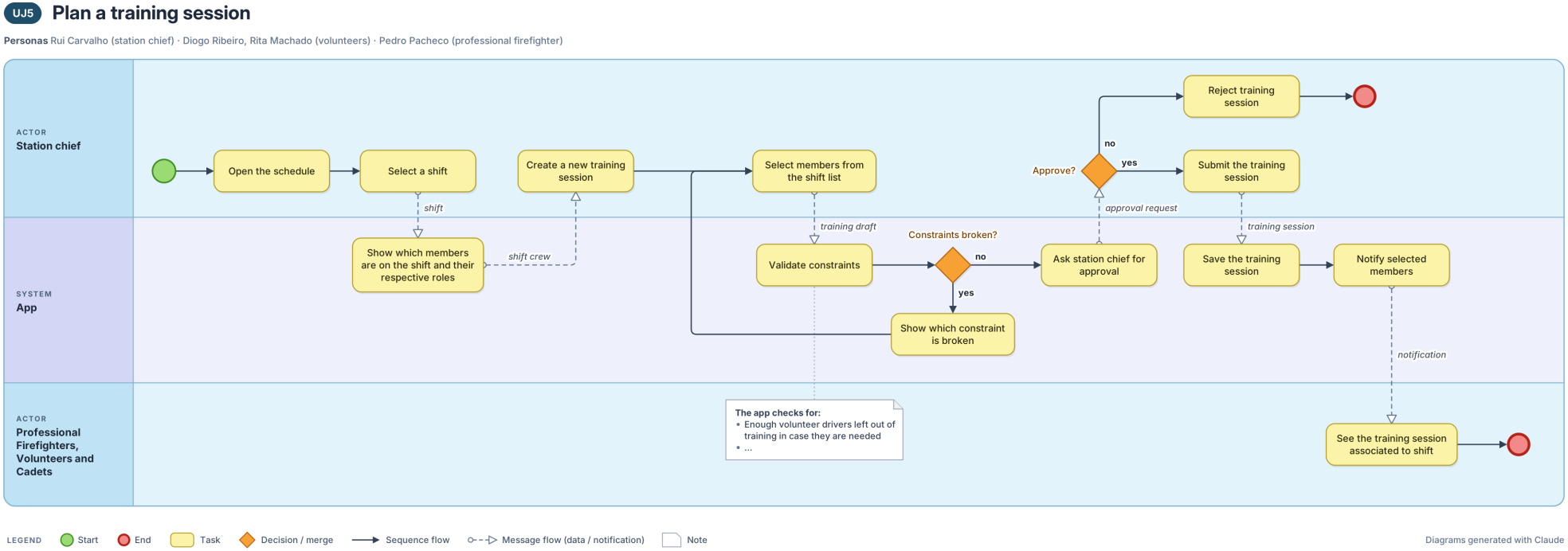


Figure 11: UJ5 – Plan a training session

6 User Workflows

Building on the user journeys, two workflows describe how Escala supports them: filling an open slot in the roster (Figure 12) and blocking an assignment when the firefighter has declared unavailability (Figure 13).

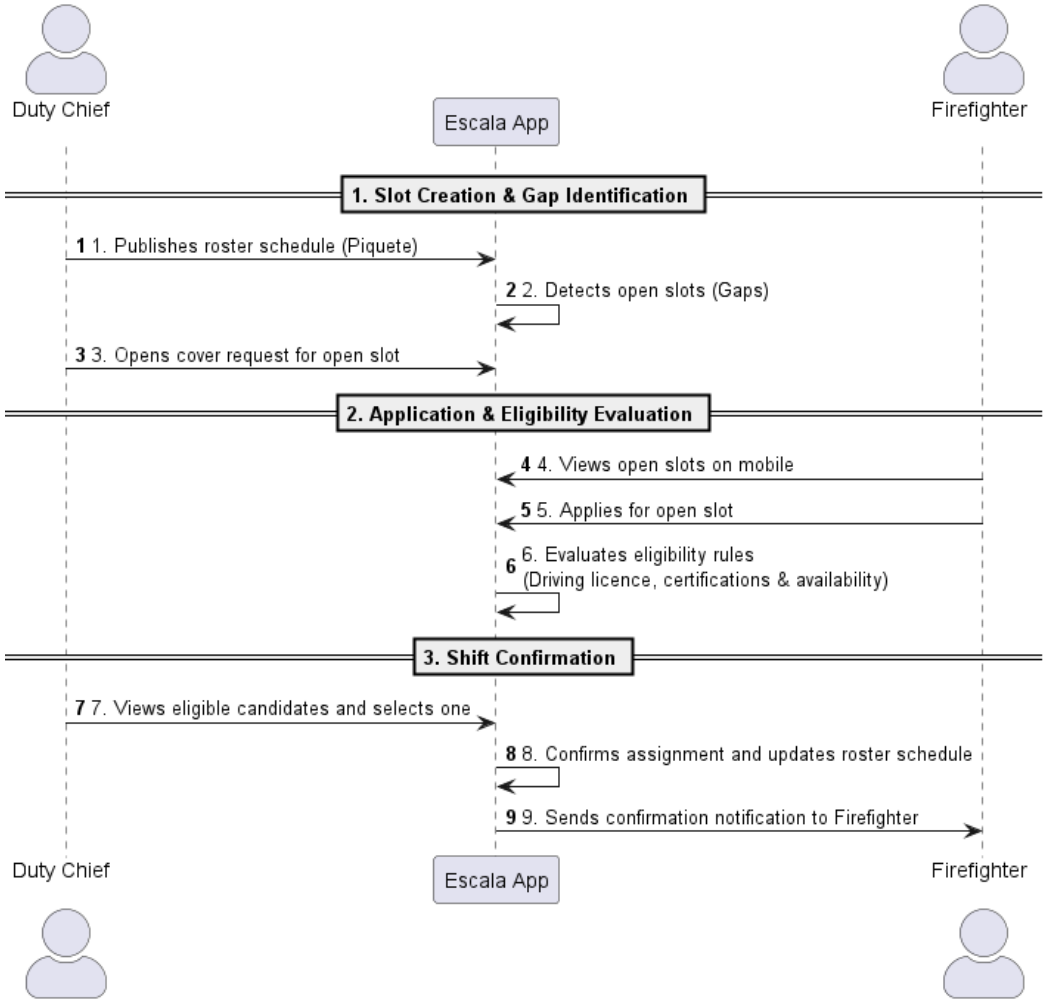


Figure 12: Workflow – Filling an open slot in the roster

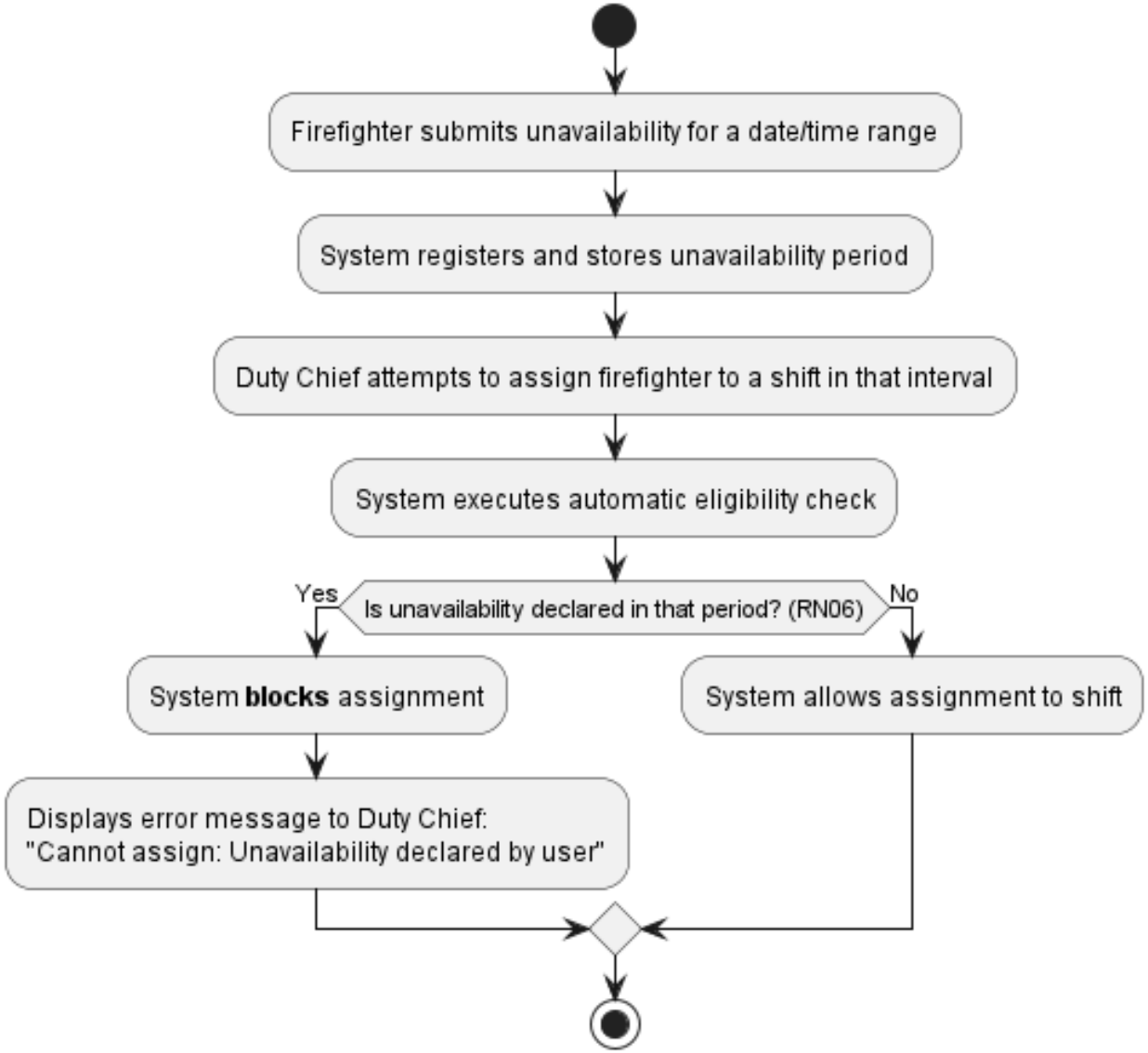


Figure 13: Workflow – Assignment blocked by declared unavailability

7 Sprint 1 – Current Status of the Deliverables

Table 9 lists the deliverables requested for the first sprint (Sprint A), grouped by phase, and the current status of each.

DELIVERABLE	STATUS	NOTES
Phase 1		
Source control Git repository setup (e.g. GitHub, Bitbucket)	**Done**	GitHub repository created.
Issue tracking system setup (e.g. Jira, GitHub)	**Done**	Jira project with epics, stories and decisions.
Agree on a team methodology	**Done**	Scrum, with a pre-game phase, sprints and a post-game phase.
Elaborate the Definition of Ready	**Done**	Documented in Confluence.
Elaborate the Definition of Done	**Done**	Documented in Confluence.
Role identification and assignment	**Done**	Product Owner, Scrum Master and Developers.
Phase 2		
Research on the project domain	**Done**	Needs of each user group described.
Research on related works	**Done**	Alternatives analysed.
Elaborate questions to customers	**In progress**	Questions prepared for the station; more personas still to be created.
Interviews with the customer (or other elicitation techniques)	**In progress**	Findings used for the personas; notes still to be documented.
Initial list of functional requirements (epics/features) and NFRs	**In progress**	Epics in Jira being aligned with the current scope.
Phase 3		
Document the functional scope proposal (product vision, user personas, epics, general workflow, domain model)	**In progress**	Product vision, personas and user journeys done; domain model drafted, to be reviewed by the team; epics being updated.
Create the Product Backlog	**In progress**	User stories in Jira being aligned with the current scope.
Prioritize the backlog with the customer (MoSCoW)	**Not started**	—
Create the Sprint B backlog (ensuring the Definition of Ready)	**Not started**	—

Table 9: Sprint A deliverables and their current status

Table 10 gives the general status of the remaining artifacts for the first delivery: 🟢 **green** complete, 🟡 **yellow** in progress and 🔴 **red** not started.

ARTIFACT	STATUS	NOTES
DoR / DoD	🟢 **Green**	Definition of Ready and Definition of Done agreed and documented in Confluence.
Team roles and work process	🟢 **Green**	Product Owner, Scrum Master and Developers assigned; Scrum process with working agreements defined.
Domain model	🟡 **Yellow**	Draft done and available in the `docs` folder of the GitHub repository; still needs a thorough review by the team.
High-level system architecture	🟢 **Green**	Done; available in the `docs` folder of the GitHub repository.

Table 10: General status of the remaining artifacts for the first delivery